Manith Marapperuma

Researcher in Artificial Intelligence | Machine Learning Engineer

Colombo, Sri Lanka ✉ manithjm@gmail.com

manithj.github.io in linkedin.com/in/manithj gh github.com/Manithj GS Scholar iD ORCID

Research Interests

Computer vision; knowledge graphs and ontology-driven reasoning; agentic and multi-agent AI systems; computational biochemistry; and quantum computing, with a focus on applying these methods to structure-based drug discovery and explainable scientific AI.

Education

General Sir John Kotelawala Defence University **2022 - 2025**
BSc (Hons) in Computer Engineering *Colombo, Sri Lanka*

- Cumulative GPA: 3.33 / 4.0 (Second Class Upper Division)
- Final Year Research Project: Grade A
- Dean's List: Semesters 7 and 8
- Relevant coursework: Bioinformatics; Machine Learning & Deep Learning; Multi-Agent Systems; Computer Vision; Natural Language Processing; High Performance Computing; Telecommunication Networks

Research & Professional Experience

Lead Research Engineer - Neuralgap **Aug 2026 - Present**
Delaware, USA (remote)

- Leading molecular dynamics simulation and engineering of viral capsids for gene therapy delivery.
- Research and development on agentic explainability, making the reasoning inside multi-step pipelines inspectable rather than opaque.
- Quantum research on molecules, applying quantum methods to molecular simulation and electronic structure.

Foundation AI Researcher - Neuralgap **Nov 2025 - Aug 2026**
Delaware, USA (remote)

- Contributed to foundation model research on protein-ligand binding affinity and pose confidence estimation, leveraging structural biology data such as docking scores, interaction geometry and pocket-level features to improve candidate triage in drug discovery.
- Built multi-engine docking coordination pipelines integrating AutoDock Vina, DiffDock and Boltz-2 for structure-based virtual screening, including pose rescoring and validation workflows.
- Designed scalable agentic architectures with memory, planning, tool-use orchestration and LLM-driven decision-making, applying foundation model reasoning across docking, rescoring and automated report generation in computational drug discovery.

Machine Learning Engineer - Augustory Corp. **May 2024 - Nov 2025**
Arizona, USA (remote)

- Built and maintained a GraphRAG-based enterprise retrieval pipeline using Neo4j, semantic search and context-aware retrieval over structured and unstructured corpora.
- Designed and optimised enterprise data-ingestion pipelines, reducing processing time by 36% and improving retrieval accuracy by 16% through algorithmic and workflow enhancements.

- Developed a real-time conversational AI agent using OpenAI Whisper and WebRTC for low-latency voice interaction, contributing across the full ML lifecycle from preprocessing and feature engineering through evaluation and deployment.

Freelance Developer - Data Science & Machine Learning **2023 - 2025**

- Delivered data science and machine learning projects for local and international clients, from requirements analysis through deployment.

Publications

1. **Marapperuma, M.**, Maduranga, M.W.P., & Damsuvi, J.K.A.T. (2026). *AI-Assisted Evaluation of Martial Arts Techniques Using Body Pose and Visual Features.* 6th International Conference on Advanced Research in Computing (ICARC). IEEE Xplore ↗

2. **Marapperuma, M.**, & Vidanage, K. (2025). *Beyond Black Boxes: An Ontology-Based Vision Framework for Transparent Fruit Disease Detection.* SLAAI International Conference on Artificial Intelligence. IEEE Xplore ↗
   Best Paper Award - AI Ontology & Knowledge Engineering Track.

3. **Marapperuma, M.**, & Vidanage, K. (2025). *Computer Vision for Object Detection in Assistive Technologies: A Comparative Review.* Department of Computer Engineering, Faculty of Computing, General Sir John Kotelawala Defence University, Sri Lanka. Paper ↗

Research Projects

Ontology-Based Vision Framework for Fruit Disease Detection

OWL, Pellet Reasoner, OpenCV, Python

- Developed a hybrid "Beyond Black Box" framework unifying a semantic knowledge base with a real-time vision pipeline for transparent fruit disease detection.
- Designed a phytopathological ontology in OWL formally codifying expert knowledge: disease symptoms, spectral phenotypes (HSV) and morphology.
- Achieved 85.00% F1 score on the public MangoFruitDDS dataset - competitive with VGG16-class deep models while remaining fully interpretable.

OntoGraph - Drug Interaction Analysis System Demo ↗

LangGraph, Semantic Web Ontologies, Open-Source LLMs, Hugging Face Spaces

- Built a graph-based biomedical reasoning system to identify and explain potential drug-drug interactions.
- Integrated LLM-driven natural language querying with semantic web ontologies for explainable insights; deployed with LangGraph orchestration and inference tracking.

SkinDet - Skin Oiliness Detection Application Demo ↗

Vision Transformers, PyTorch, Streamlit

- Fine-tuned pretrained ViT weights on a custom synthetic skincare dataset, reaching 92% test accuracy for skin oiliness classification from facial images.
- Deployed an interactive Streamlit application for real-time inference and visualisation.

Fine-Tuned Gemma-2B for English-Sinhala Translation Model ↗

Hugging Face, PyTorch, LoRA / QLoRA, PEFT

- Fine-tuned Google Gemma-2B with LoRA and QLoRA for efficient English-Sinhala translation, using PEFT to train within a single-GPU memory budget.
- Instruction-tuned on a Sinhala-English dataset for improved contextual accuracy; released publicly on the Hugging Face Hub.

Awards & Honours

- Best Paper Award, AI Ontology & Knowledge Engineering Track, SLAAI International Conference on Artificial Intelligence (2025)
- Best Technology in AI Award, Data Odyssey 2024
- Shubhra Kar Linux Foundation Training (LiFT) Scholarship Recipient
- IEEEXtreme 17.0 Global Programming Competition - Island Rank 10
- IEEEXtreme 16.0 Global Programming Competition - Island Rank 40
- Finalist, Genesiz 2024 Inter-University Robotics Competition
- Finalist, Cypher 2.0 Capture the Flag (CTF) Hackathon Challenge
- Runner-up, BleedCode 2.0 Intra-Faculty Programming Competition
- President Scout Award, World Organization of the Scout Movement

Invited Talks & Workshops

- **Guest Speaker**, IEEEXtreme 19.0 Workshop - designed and delivered a technical workshop on "Utilizing Generative AI to Accelerate Problem Solving." LinkedIn ↗
- **Guest Speaker**, IEEEXtreme 18.0 Awareness Session - co-presented team strategies behind a top-10 national ranking. LinkedIn ↗

Technical Skills

Agentic AI	Multi-agent orchestration, LangGraph, LangChain, MCP, prompt engineering, LLMs
Retrieval	RAG, GraphRAG, Neo4j, semantic search, FAISS, Pinecone, Qdrant
ML / DL	PyTorch, TensorFlow, Scikit-learn, OpenCV, Vision Transformers, LoRA/QLoRA/PEFT
Comp. biochem	AutoDock Vina, DiffDock, Boltz-2, GNINA, DynamicBind, molecular dynamics pipelines
Languages	Python, SQL, C/C++ (CUDA)
Infrastructure	Docker, Kubernetes, AWS (Bedrock, ECS, EKS, S3, CloudWatch), MLflow, FastAPI, Flask, Git, Linux
Data stores	PostgreSQL, MySQL, MongoDB

Certifications

- NVIDIA - Generative AI with Diffusion Models
- NVIDIA - Accelerated Computing in CUDA C/C++
- Oracle - Generative AI Professional
- DeepLearning.AI - Machine Learning Specialization; Deep Learning Specialization; TensorFlow Developer Professional
- AWS Certified Cloud Practitioner (in progress)

Service & Leadership

Webmaster, IEEE Student Branch of KDU **2024 - 2025**
Volunteer Affairs Manager, IEEE Young Professionals Sri Lanka **2023 - 2025**
Member, AI & Data Science Club of KDU **2023 - 2025**